Practical-01

AIM- Study the Data Analytics lifecycle and perform data acquisition from CSV, Excel, and JSON les using Pandas.

Step 1: import the pandas library

```
#Import the pandas library
import pandas as pd
```

Step 2: create sample **data**

```
#Create sample data
data = {
    "ID" : [101,102,103,104,105,106,107],
    "Name" : ["Rahul","Priya","Amit","Riya","Karan","Krishna", "Harry"],
    "Age" : [20,21,19,20,22,24,25],
    "City" : [ "surat","navsari","valsad","vapi","bhruch","rajkot","surat"],
    "marks": [85,86,84,82,83,84,86]
}
```

Step 3: create a dataframe

```
# Step 3: Create a DataFrame

df = pd.DataFrame(data)
df
```

	ID	Name	Age	City	marks
0	101	Rahul	20	surat	85
1	102	Priya	21	navsari	86
2	103	Amit	19	valsad	84
3	104	Riya	20	vapi	82
4	105	Karan	22	bhruch	83
5	106	Krishna	24	rajkot	84
6	107	Harry	25	surat	86

Step 4: save data as a csv

```
# Step 3: Create a DataFrame
df.to_csv("students.csv", index=False)
print("CSV file created successfully.")
```

```
CSV file created successfully.
```

Step 5: save data as an excel file

```
# Step 5: Save the DataFrame as an Excel file
df.to_excel("students.xlsx", index=False)
print("Excle file created successfully.")
```

```
Excle file created successfully.
```

Step 6: save the data as a JSON file

```
# Step 6: Save the DataFrame as a JSON file
df.to_json("students.json" , orient="records", indent=4)
print("JSON file created successfully.")
```

```
JSON file created successfully.
```

Step 7: Read The CSV File

```
# Step 7: Read the CSV file
df_csv = pd.read_csv("students.csv")
```

```
df_csv
```

	ID	Name	Age	City	marks
0	101	Rahul	20	surat	85
1	102	Priya	21	navsari	86
2	103	Amit	19	valsad	84
3	104	Riya	20	vapi	82
4	105	Karan	22	bhruch	83
5	106	Krishna	24	rajkot	84
6	107	Harry	25	surat	86

Step 8 : Read The Excle File

```
# Step 8: Read the Excel file
df_excel = pd.read_excel("students.xlsx")
df_excel
```

	ID	Name	Age	City	marks
0	101	Rahul	20	surat	85
1	102	Priya	21	navsari	86
2	103	Amit	19	valsad	84
3	104	Riya	20	vapi	82
4	105	Karan	22	bhruch	83
5	106	Krishna	24	rajkot	84
6	107	Harry	25	surat	86

Step 9: read the JSON file

```
# Step 9: Read the JSON file
df_json = pd.read_json("students.json")
df_json
```

	ID	Name	Age	City	marks
0	101	Rahul	20	surat	85
1	102	Priya	21	navsari	86
2	103	Amit	19	valsad	84
3	104	Riya	20	vapi	82
4	105	Karan	22	bhruch	83
5	106	Krishna	24	rajkot	84
6	107	Harry	25	surat	86

Step 10: display the first five record

```
# Step 10: Display the first five records
df_csv.head()
```

	ID	Name	Age	City	marks
0	101	Rahul	20	surat	85
1	102	Priya	21	navsari	86
2	103	Amit	19	valsad	84
3	104	Riya	20	vapi	82
4	105	Karan	22	bhruch	83

Step 11: Display the last five record

```
# Step 11: Display the last five records
df_csv.tail()
```

	ID	Name	Age	City	marks
2	103	Amit	19	valsad	84
3	104	Riya	20	vapi	82
4	105	Karan	22	bhruch	83
5	106	Krishna	24	rajkot	84
6	107	Harry	25	surat	86

Step 12 : Display dataset information

```
# Step 12: Display dataset information
df_csv.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   ID      7 non-null      int64 
 1   Name    7 non-null      object
 2   Age     7 non-null      int64 
 3   City    7 non-null      object
 4   marks   7 non-null      int64 
dtypes: int64(3), object(2)
memory usage: 412.0+ bytes
```

Step 13 :display statistical Summary

```
# Step 13: Display the statistical summary
df_csv.describe()
```

	ID	Age	marks
count	7.000000	7.000000	7.000000
mean	104.000000	21.571429	84.285714
std	2.160247	2.225395	1.496026
min	101.000000	19.000000	82.000000
25%	102.500000	20.000000	83.500000
50%	104.000000	21.000000	84.000000
75%	105.500000	23.000000	85.500000
max	107.000000	25.000000	86.000000

Step 14 :Display Dataset shape

```
# Step 14: Display the dataset shape
df_csv.shape
```

```
(7, 5)
```

Step 15 : Display Columns Names

```
df_csv.columns
```

```
Index(['ID', 'Name', 'Age', 'City', 'marks'], dtype='object')
```

Step 16: Display Data Types

```
df_csv.dtypes
```

	0
ID	int64
Name	object
Age	int64
City	object
marks	int64

dtype: object

Step 17: Check For Missing values

```
df_csv.isnull().sum()
```

	0
ID	0
Name	0
Age	0
City	0
marks	0

dtype: int64

Step 18 : Display a Single Column

```
df_csv["marks"]
```

	marks
0	85
1	86
2	84
3	82
4	83
5	84
6	86

dtype: int64

Step 19 :Display muitiple column

```
df_csv[["marks" , "Name"]]
```

	marks	Name
0	85	Rahul
1	86	Priya
2	84	Amit
3	82	Riya
4	83	Karan
5	84	Krishna
6	86	Harry

Step 20 : Filter Record

```
df_csv[df_csv["marks"] > 85]
```

	ID	Name	Age	City	marks
1	102	Priya	21	navsari	86
6	107	Harry	25	surat	86

Step 21: calculate Average Marks

```
df_csv["marks"].mean()
```

```
np.float64(84.28571428571429)
```

Step 22 : Sort Data

```
df_csv.sort_values(by="marks",ascending=False)
```

	ID	Name	Age	City	marks
1	102	Priya	21	navsari	86
6	107	Harry	25	surat	86
0	101	Rahul	20	surat	85
2	103	Amit	19	valsad	84
5	106	Krishna	24	rajkot	84
4	105	Karan	22	bhruch	83
3	104	Riya	20	vapi	82